

That's a whole lot of tasks, so it makes sense that it got the name 'multi'-meter'. This tool is useful for basic sanity checks and troubleshooting as well. In our tutorial, we will specifically focus on measuring voltage, current, resistance and continuity.

Parts of a multimeter

A multimeter traditionally consists of three parts - display, selection knob and ports. Let's explore each one of these elements separately.

The display on a multimeter has four digits usually and can display a negative sign for when you might have connected the leads or probes to the terminals with opposite polarity. There are also variants which have illuminated displays for low-light situations and there are ones with a lot more than 4 digits but they obviously cost more.

The selection knob lets users set the multimeter specifically to read things such as milliamps or amperes (mA) of current, voltage (V), capacitance, and resistance (Ω).

The ports on a multimeter have two probes that are plugged into them on the front of the unit. The COM probe, which stands for common and is usually connected to Ground or '-' of a circuit, is usually black. A 10A port (or a higher value) is used to measure large currents. The mA or mV or V or Ω port is usually the one where the red probe is connected and measures current, voltage and resistance. These ports aren't all standard and vary from model to model, simply refer to the user manual to know exactly which ports to use for which range of the parameter that you are testing.

If you don't want to use probes, then there are banana plugs which are sort of a clip-and-forget connector. So you can free your hands to do other things while monitoring the values on the multimeter.

Measuring voltage

Let's begin with measuring the voltage of a simple AA battery. To do this, you must plug the black probe into the COM and

